$f(X) \in \mathbb{R}^d$ Laplace $\left(0, \frac{\Delta}{\epsilon}\right)$ in all components independently of each other.

$$\Delta = \max_{x, x' \text{ neighboring}} \| f(x) - f(x') \|_1 = \sum_{i=1}^{d} | (f(x))_i - (f(x'))_i |$$

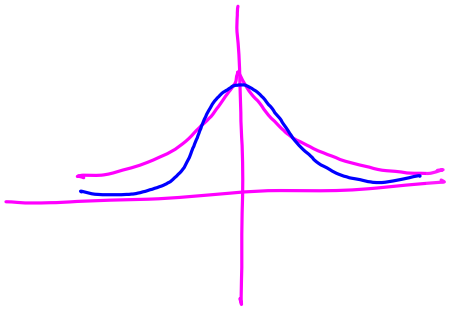
Approximate DP

$M : \mathcal{X}^n \to \mathcal{Y}$ is (ϵ, δ) approximately DP if $\forall$ neighbouring $x, x' \in \mathcal{X}^n$, and all $S \subseteq \mathcal{Y}$

$$P(M(x) \in S) \leq e^{\epsilon} P(M(x') \in S) + \delta$$

To achieve this, use the Gaussian Mechanism

Add Gaussian noise $N\left(0, \ln\left(\frac{1}{\delta}\right) \frac{\Delta^2}{\epsilon^2}\right)$



For high dimensional outputs,

$$N\left(0, \ln\left(\frac{1}{\delta}\right) \frac{\Delta_2^2}{\epsilon^2}\right)$$ L2- sensitivity.

$$\Delta_2 = \max_{x, x' \text{ neighbouring}} \|f(x) - f(x')\|_2 = \sqrt{\sum_{i=1}^{d} (f(x)_i - f(x')_i)^2}$$

We gain in terms of utility

Laplacian noise $\quad Lap\left(0, \frac{\Delta}{\varepsilon}\right) \qquad Lap\left(0, \frac{d}{n\varepsilon}\right)$

Gaussian noise $\quad N\left(0, \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}\right)$

$\sum_{i=1}^{d} \frac{1}{n} = \frac{d}{n}$

Properties of ADP

- POST PROCESSING

Say M is (ϵ, δ) DP

$F \circ M$ is (ϵ, δ) DP

- Composition

$\{M_1, M_2, \ldots, M_k\}$ are all (ϵ, δ) DP

Basic: $(k\epsilon, k\delta)$ DP

Advanced: $\left(\epsilon\sqrt{k\log\left(\frac{1}{\delta'}\right)} + \epsilon\left(e^{\epsilon}-1\right), \; k\delta + \delta'\right)$ - DP

NPTEL

Digital good



Buyers n

Valuations $v_1, \ldots, v_n$

How to set price?

p

$p\left(\sum_{i=1}^{n} \mathbb{1}(v_i \geq p)\right)$

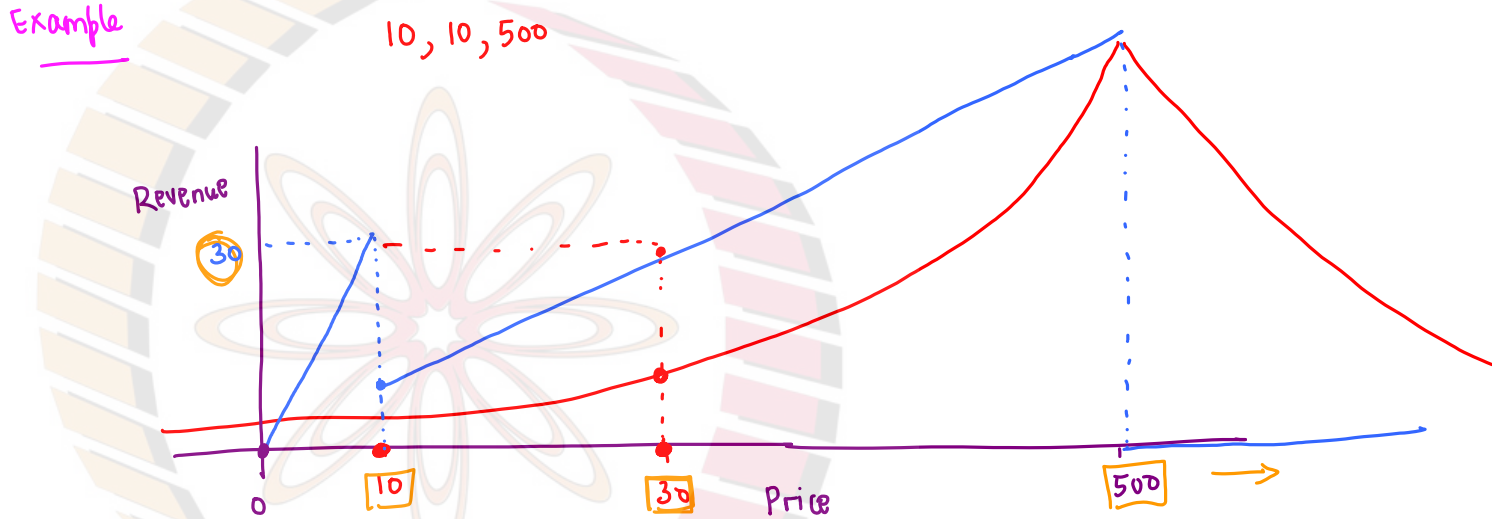
Indicator

↳ # Buyers whose valuation $\geq p$.

NPTEL



Example



NPTEL

EXPONENTIAL MECHANISM (2012, Talwar et.al)

$x \in X^n$ (valuations) (10, 10, 500)

$\mathcal{H}$ (set of prices)

$S: (X^n, \mathcal{H}) \to \mathbb{R}$ Revenue function.

$$p \sum_{i=1}^{n} \mathbb{I}(v_i \geq p)$$

$$\Delta = \max_{h \in \mathcal{H}} \max_{x, x' \in X^n} |S(x, h) - S(x', h)|$$

Select $h \in \mathcal{H}$ with probability proportional to $e^{\frac{\epsilon}{2\Delta} S(x, h)}$

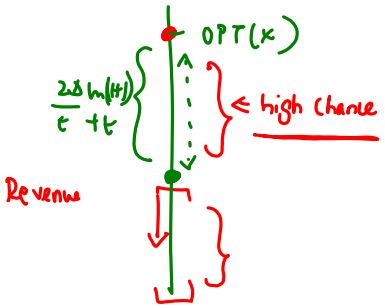
EM is ϵ-DP

NPTEL

Utility

$$P\Big(S(x, EM(x)) \le OPT(x) - \Big(\frac{2\Delta}{\epsilon}\ln\Big(\frac{1}{\gamma t}\Big) + t\Big)\Big) \le e^{-t}$$

- $S(x, EM(x))$: Price → Revenue
- $OPT(x)$: Maximum revenue that could be achieved.



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